Examiner only

6. *(a)* Fertilisers usually contain compounds of three essential elements required for healthy and productive plant growth. The table shows the properties of two nitrogenous fertilisers.

Property	Ammonium nitrate	Urea
% nitrogen	34	46
Mass of fertiliser absorbed by plants per 2000 litre spreader (kg)	690	560
Solubility	very soluble	very soluble
Cost (£/tonne)	360	350
Loss of ammonia due to evaporation	low	high
Weather dependency	effective in all conditions	effectiveness dependent on specific temperature and rainfall conditions

(i) Which fertiliser leads to the absorption of more **nitrogen** from a full 2000 litre spreader? Show your working. [2]

Fertiliser ....

(ii) Urea is the most widely used fertiliser in the world. However, many British farmers prefer to use ammonium nitrate.

Put a tick (✓) in the box next to the statement which suggests the **main** reason for this. [1]

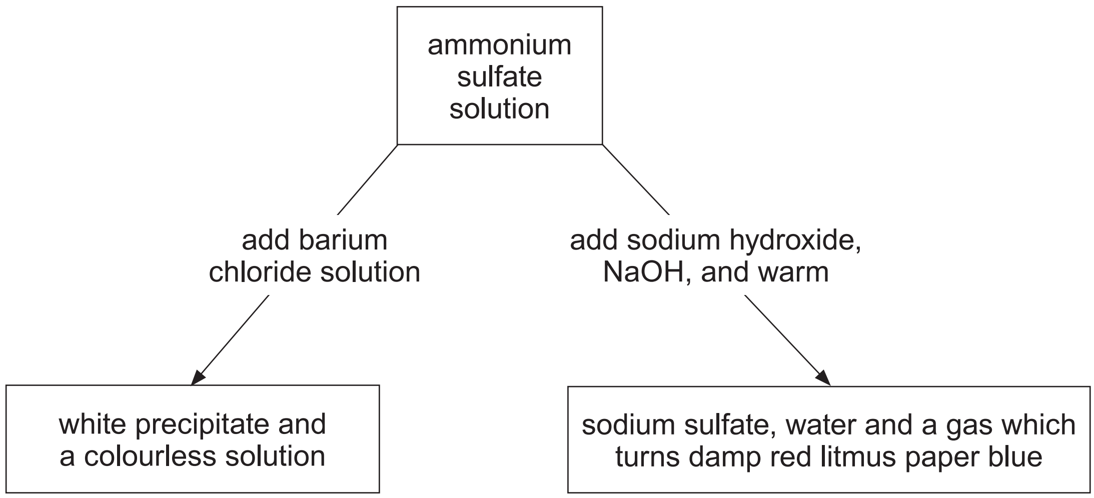
ammonium nitrate contains less nitrogen than urea ☐

ammonium nitrate is better suited to British weather conditions than urea ☐

more ammonium nitrate is absorbed by plants than urea ☐

ammonium nitrate is more expensive than urea ☐

(b) The flow chart shows the chemical tests for the identification of the ions present in ammonium sulfate, $(NH_4)_2SO_4$.



(i) Write a balanced **symbol** equation for the reaction between ammonium sulfate and sodium hydroxide. [2]

....

(ii) The symbol equation below represents the reaction occurring between ammonium sulfate solution and barium chloride solution.

$$BaCl_2(aq) \quad + \quad (NH_4)_2SO_4(aq) \longrightarrow 2NH_4Cl(aq) \quad + \quad BaSO_4(s)$$

Write the **ionic** equation for the reaction. Include the state symbols. [2]

.... + ... $\longrightarrow$ ....

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